1 Overview

1.1. Informal description

The “Biomass Inventory Mapping and Analysis Tool Business Data” series is a collection of datasets that are related to the supply and location of biomass in Canada. It includes data about leftover material from agricultural and forestry industries (residues), agricultural crops, and municipal solid waste that can be used as feedstock for bioindustries.

The data is used by a web-based application, the Biomass Inventory Mapping and Analysis Tool or BIMAT, to allow users to view and analyze detailed information about biomass availability using digital maps and database searches. Users can search by type of biomass, by required quantity and by location. Users also have the option to include in their reports estimates of harvest and transportation costs and information about 1-in-10 year low and 1-in-20 year low biomass production.

1.2. Data product specification metadata

This section provides metadata about the creation of this data product specification.

Data product specification – title:	BIMAT – Business Data
Data product specification – reference date:	2019-10-04
Data product specification – responsible party:	Agri-Geomatics
Data product specification – language:	English, French
Data product specification – topic category:	Farming, Economy, Transportation

1.3. Terms and definitions

- **Biomass**

  Living and recently living biological material that can be used for industrial production.

- **Crop residue**

  The non-grain above-ground portion of a crop plant is left behind after harvest. It is composed of three parts:

- **Chaff**

  Seed coverings and other debris separated from the seed by threshing at time of harvest.

- **Straw**

  The dry coarse stems of crop plants separated from the grain, chaff and stubble during harvest.

- **Stubble**

  The short standing base of stems remaining on a field after harvest.

- **Crop yield and production**

  Yield is the amount of grain or residue produced per unit of land area (e.g.. hectare, acre) while production is the amount produced in a geographic area (e.g. field, BIMAT grid cell, Statistics Canada reporting area). The production is yield multiplied by the land area.

- **Conventional tillage**

  A tillage system that uses cultivation (inverting the soil) as a means to prepare the soil for seeding and control weeds.

- **Feedstock**

  Raw material used in the manufacture of a product or an industrial process. When discussing biomass, feedstock is the biological raw material for end products such as biodiesel, ethanol, or fibre board.

- **Zero Till**

  A method of planting crops with minimal disturbance of the soil surface. This is sometimes referred to as no-till farming or direct seeding.

1.4. Abbreviations

BIMAT - Biomass Inventory Mapping and Analysis Tool

IBSAL - Integrated Biomass Supply Analysis and Logistics

NDVI - Normalized Difference Vegetation Index

NRN - National Road Network

2. SPECIFICATION SCOPE

This data specification has only one scope, the general scope.

NOTE: The term ‘specification scope’ originates from the International Standard ISO19131.

‘Specification scope’ does not express the purpose for the creation of a data specification or the potential use of data, but identifies partitions of the data specification where specific requirements apply.

4.2.1.69. Total Residential Municipal Solid Waste

Name	MNCPL_SOLID_WASTE_TOTAL_VOL (Total Residential Municipal Solid Waste Volume)		
Definition	Total Residential Municipal Solid Waste in tonnes, calculated for each BIMAT grid cell within a population centre.		
Aliases			
Producer	National Research Council Canada		
Value Data Type	Double		
Value Domain Type	0 (not enumerated)		
Value Domain			
	Feature Attribute Value		
	Label	Code	Definition

4.2.1.70. Total Organic Waste

Name	MNCPL_SOLID_WASTE_ORGANIC_VOL (Total Organic Waste Volume)		
Definition	Total Organic Waste (including Food and Yard waste) in tonnes, calculated for each BIMAT grid cell within a population centre. This includes both diverted and non-diverted waste quantities.		
Aliases			
Producer	National Research Council Canada		
Value Data Type	Double		
Value Domain Type	0 (not enumerated)		
Value Domain			
	Feature Attribute Value		
	Label	Code	Definition

4.2.1.71. Total Paper Waste

Name	MNCPL_SOLID_WASTE_PAPER_VOL (Total Paper Waste Volume)		
Definition	Total Paper Waste in tonnes, calculated for each BIMAT grid cell within a population centre. This includes both diverted and non-diverted waste quantities.		
Aliases			
Producer	National Research Council Canada		
Value Data Type	Double		
Value Domain Type	0 (not enumerated)		
Value Domain			
	Feature Attribute Value		
	Label	Code	Definition